

GridSense

Grid-Aware Energy Copilot for Victorian Households

1. Startup Pitch

Problem

Modern electricity grids—especially in regions like Victoria—are becoming unstable and inefficient due to:

- High rooftop solar causing midday oversupply
- Evening peak demand stressing infrastructure
- Increasing electricity costs for consumers
- Lack of real-time decision-making tools for households

Consumers today:

- Do not know when to use electricity
 - Cannot respond to grid conditions
 - Unintentionally increase costs and grid stress
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Solution

Grid-Aware Energy Copilot

A software platform that:

- Analyzes household energy usage
 - Predicts grid conditions (price, demand, solar availability)
 - Provides real-time actionable recommendations
 - Automates appliance usage for optimal efficiency
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Core Value Proposition

For users:

- Reduce electricity bills
- Automate smart energy usage
- No technical knowledge required

For the grid:

- Smooth demand peaks
 - Improve renewable utilization
 - Enable decentralized demand response
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Product Roadmap

Phase 1:

- Upload smart meter data (NEM12)
- Get personalized energy recommendations

Phase 2:

- Real-time insights using connected devices

Phase 3:

- Automated energy control via smart plugs

Phase 4:

- Network-level coordination (Virtual Demand Response)
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How It Works

1. Collect user energy data
 2. Combine with weather, grid demand, and pricing signals
 3. Predict cheapest and most efficient usage times
 4. Optimize appliance schedules
 5. Deliver real-time actionable recommendations
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Market Opportunity

- High rooftop solar penetration in Australia
 - Increasing adoption of EVs, batteries, and smart homes
 - Growing need for demand-side energy coordination
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Why Now

- Rapid renewable adoption
- Rising electricity prices

- Increased smart device penetration
 - Policy push toward demand-side participation
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Business Model

Freemium Model:

- Free: Basic insights
- Paid: Advanced optimization and automation

Future Revenue Streams:

- Utility partnerships
 - Demand response programs
 - API licensing
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Competitive Advantage

Most solutions provide data visualization.

GridSense provides decision-making and automation.

Vision

To become the operating system for household energy usage in distributed renewable grids.

2. Product Development Plan

Stage 1: MVP (0–6 weeks)

- NEM12 parser
- Usage visualization
- Rule-based recommendations

Stage 2: Intelligence Layer (6–12 weeks)

- Demand forecasting
- Tariff modeling
- Weather integration
- Optimization engine

Stage 3: Decision Engine (12–20 weeks)

- Scheduling optimizer
- Cost minimization

Stage 4: Real-Time System

- Integration with smart plugs or CT sensors
- Live monitoring

Stage 5: Automation Layer

- IoT integration
- Autonomous energy decisions

Stage 6: Network Effects

- Aggregate users
 - Build virtual demand response network
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3. Technical Architecture

Backend

- FastAPI (Python)
- Data ingestion, ML, optimization

Machine Learning

- Demand forecasting
- Solar prediction
- Grid stress modeling

Optimization

- OR-Tools / PuLP

Frontend

- React / Next.js

Data Storage

- PostgreSQL / InfluxDB

Future

- IoT integration via MQTT

4. Go-To-Market Strategy

Step 1

Target early adopters:

- Solar owners
- EV users
- Tech-savvy households

Step 2

Deliver clear value:

- Reduce electricity costs

Step 3

Build trust:

- Simple UI
- No hardware requirement initially

Step 4

Expand:

- Add automation
- Partner with installers

5. Key Metrics

- Electricity cost reduction
- Peak usage reduction
- User retention
- Recommendation adoption rate

6. Risks and Mitigation

Risks

- Low user engagement
- Limited data access

- Hardware adoption barriers

Mitigation

- Simplify user experience
 - Focus on automation
 - Start software-first
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7. Strategic Positioning

GridSense is not a monitoring tool.

It is a decision-making system for energy optimization in modern distributed grids.